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Title: Correlation between dissolved oxygen concentrations and FeMn nodule geochemistry in the Samoa Basin

Abstract

Ferromanganese nodules are a wide-spread oceanographic mineral formation that are found in abyssal plains and around hydrothermal vents. Despite their prevalence, their remoteness and complexity of their formation means that the relationship they have with their environments is not entirely known. The project aims to address one aspect of nodule environments by finding a correlation between dissolved oxygen in abyssal sediments and nodule abundance and geochemistry. This data comes from the Samoa Basin, which is a particularly important mineral formation region due to a possible lease sale of the area for future mining operations. Identifying connections between the nodules and their environments will hopefully better inform the public about how mining could impact the wider environment and demonstrate the role that the nodules have on global ocean oxygen cycles.

Introduction

Ferromanganese nodules (FeMn nodules, nodules) occur across the world oceans, and recent attention has been paid to them as a resource for critical metals (Petersen et al., 2016). Nodules are mineral deposits that form mostly on the sediment surface at water depths below 4000 meters where there is a low sedimentation rate. They are made up of Fe and Mn oxides in a variety of mineral phases, and also contain high levels of trace metals and rare earth elements (REY) (Wu et al., 2022). Nodule formations are extraordinarily variable, ranging from sub-mm to 21 cm in diameter, and they form a variety of shapes and contain variable growth structures (Kuhn, 2017). The variability of nodules can be attributed to the ways that they form at the sea floor. Nodules are either formed hydrogenetically from metals precipitating directly out of sea water, or diagenetically from metals precipitating out of porewater in sediments. Hydrogenetic nodules have higher concentrations of Fe, Co, and REY compared to digenetic nodules, and they have Mn/Fe ratios of less than 2.5, while diagenetic nodules have higher concentrations of Mn, Ni, Cu, and Mn/Fe ratios of more than 4 (Menendez et al., 2019). Whether a nodule is diagenetic or hydrogenetic depends on environmental conditions, including the oxygenation of the seawater they form in; hydrogenetic nodules form in more oxygenated conditions, while diagenetic nodules form in sub-oxic conditions. Thus, quantifying environment conditions such as dissolved oxygen concentrations is vital to understanding their formation and geochemical makeup. Dissolved oxygen in sediments is stored in pore water and can be extraordinarily variable across the ocean. It is controlled by the conditions both of the overlying water column, including water temperature, biological production at the surface, and water age (Smith Jr. et al., 2022), and the conditions of the sediments itself including the amount of organic matter deposited into the sediments, the amount of remineralizing microbes in the sediment, and the physical characteristics of the sediments such as density and grainsize (Jørgensen et al., 2022). This project aims to connect dissolved oxygen concentrations in sediment to the abundance and basic geochemical makeup of nodules from data collected from the Samoa Basin.

Hypotheses and Method plan

I pose several hypotheses about the connection between dissolved oxygen in sediments hosting FeMn nodules:

Hypothesis 1) Geochemistry of nodules will reflect mostly hydrogenetic formation across all sampling locations.

Hypothesis 2) Sediment with lower dissolved oxygen will occur in locations with higher total nodule abundance.

Hypothesis 3) Sediment from sampling locations with nodules containing lower Mn/Fe ratios will have higher dissolved oxygen concentrations.

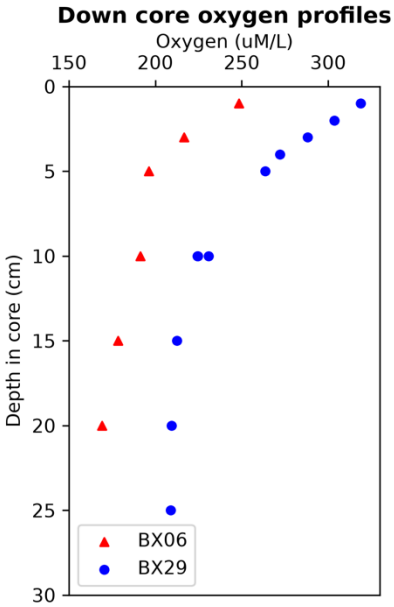


Figure 1: Profile of dissolved oxygen from two sediments cores. BX06 had the lowest surface oxygen concentrations from the dataset and BX29 had the highest surface oxygen concentrations.

Data to use:

To address my hypotheses, I will utilize data from samples collected in 2026 from the Samoa Basin. This data is made up of three main data types: oxygen data (Shapiro et. al. 2026), nodule abundance data, and nodule geochemistry (Adamczyk et. al., 2026). The oxygen data spans across 34 locations in the study region, and includes ten to twenty data points down core depending on the core length and granularity of measurements. Two downcore profiles are shown in Figure 1 and show the variability of dissolved oxygen data in the region, with BX06 as the lowest dissolved surface oxygen concentrations and BX29 as the highest surface oxygen concentration. The nodule abundance data reports nodule wet weight in kilograms per meter for a total of 35 locations, and the spread of this data is shown in Figure 2. There are 11 locations with less than 0.5 kg/m^2 of FeMn nodules, and the maximum weight was 21.0 kg/m^2. The nodule geochemistry data contains element concentrations from a representative sample from each location and includes data for major and minor elements as well as rare earth elements in parts per million. The timeline of this project is shown in Figure 3. Data collection and organization will occur in weeks one and two, as well as the quality control and cleaning as described below.

Quality control:

Data used in this study has undergone its own quality check, so data will be assumed to be accurate where reported. Oxygen microprofile data reports quality control codes, so only data with 100% confidence will be used. Data that is below detection limit will be treated at zero or N/A values to allow for easier graphical and statistical representation of the data. Where there have been duplicates collected in the oxygen data or the geochemical data, the average between the two values will be used, and I will report a standard deviation to ensure one value was not erroneously large or small.

Expected analysis:

To analyze my data and draw connections between dissolved oxygen and nodules in the Samoan basin, I will use various python tools and packages such as *SciPy* and *Scikit-learn*. I will first collect some summary statistics on the nodule geochemistry data and compute a Fe/Mn ratio for each nodule and an average for each sample location. I will also compare other major elements that are known to be linked with environmental conditions, such as Co and Cu. Once I determine the relationships between different elements across sample locations, I will create a series of graphs and charts to capture any trends. I will also compare the organized geochemical data to widely available hydrogenetic and diagenetic end-member geochemical values, which will help me address hypothesis 1. I will then relate the geochemical data

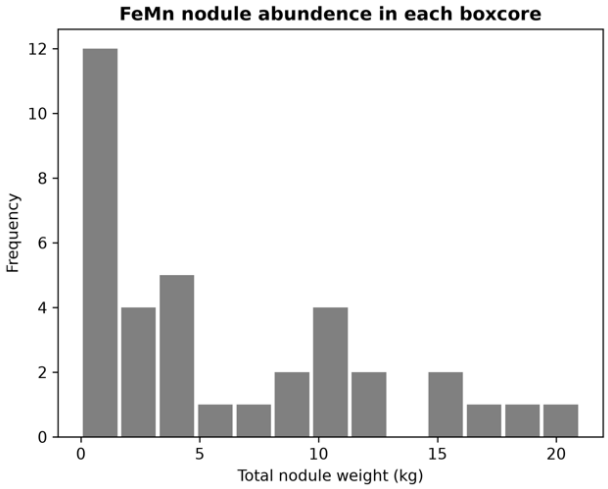


Figure 2: Histogram of nodule abundance from the dataset.

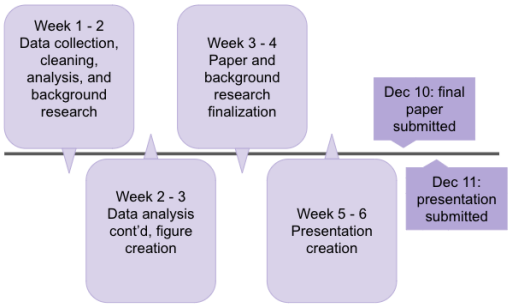


Figure 3: Timeline for project starting from the first week of November

to the surface oxygen data to see if there is a relationship between any geochemical trends and the sediment oxygen that the nodules resided in to address hypothesis 3. To address hypothesis 2, I will look for trends between nodule abundance and oxygen data by comparing statistics and the geographical spread of the data. All the relationships and trends that I find between geochemistry and oxygen I will use a correlation coefficient such as the r-value to determine if any relationship is statistically significant. Statistical summaries and graphical representations will be created during weeks two and three, as shown in the Figure 3 timeline.

If relationship is measured between any of the variables, I will also attempt to create a model that will estimate one parameter given the other; for example, I hope to estimate the nodule abundance given sediment oxygenation, or a specific Fe/Mn ratio based on oxygen data if there is a relationship between these two datasets. This will occur during week 3 and 4, if applicable, while the paper and background research is also finalized.

Broader impacts

This area of research is particularly important because of mining interest. On November 19, 2026, an auction is expected to take place for the region where these samples were collected (Marine Minerals Administration). Although the lease sale does not authorize immediate mining, it allows mapping and further data collection. Thus, understanding these environments and the relationship nodules might have with their surrounding sediment is vital to inform how mining operations might affect the area.

References

Jørgensen, B. B., Wenzhöfer, F., Egger, M., & Glud, R. N. (2022). Sediment oxygen consumption: Role in the global marine carbon cycle. *Earth-Science Reviews*, *228*, 103987. https://doi.org/10.1016/j.earscirev.2022.103987

Kuhn, T., Wegorzewski, A., Ruhlemann C., Vink A.. (2017). Composition, Formation, and Occurrence of Polymetallic Nodules. In Sharma, R., (Ed) *Deep-Sea Mining*, (pp. 23 - 64). Springer.

Marine Minerals Administration. (2026, October 19). *American Samoa activities*. U.S. Department of the Interior. https://www.boem.gov/marine-minerals/american-samoa-activities

Menendez, A., James, R. H., Lichtschlag, A., Connelly, D., & Peel, K. (2019). Controls on the chemical composition of ferromanganese nodules in the Clarion-Clipperton Fracture Zone, eastern equatorial Pacific. *Marine Geology*, *409*, 1–14. https://doi.org/10.1016/j.margeo.2018.12.004

Petersen, S., Krätschell, A., Augustin, N., Jamieson, J., Hein, J. R., & Hannington, M. D. (2016). News from the seabed – Geological characteristics and resource potential of deep-sea mineral resources. *Marine Policy*, *70*, 175–187. https://doi.org/10.1016/j.marpol.2016.03.012

Smith Jr., K. L., Messié, M., Connolly, T. P., & Huffard, C. L. (2022). Decadal Time-Series Depletion of Dissolved Oxygen at Abyssal Depths in the Northeast Pacific. *Geophysical Research Letters*, *49*(24), e2022GL101018. https://doi.org/10.1029/2022GL101018

Wu, H., Ge, C., Cai, Y., Shi, X., & Huang, M. (2022). Structure, composition, and formation of ferromanganese micronodules at the eastern southwest Indian ridge. *Marine Geology*, *451*, 106874. https://doi.org/10.1016/j.margeo.2022.106874